

A Pre-feasibility study for the establishment of a children's museum in Aqaba



2017



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1. Executive summary

The idea of the project summed up in establishing a specialized museum for children, as it aims to be a place to the families of the region or tourists.

The idea was presented for the lack of the existence of such museums in AlAqaba, despite the availability of the need for such projects. Based on the analysis of the results of the proposed project, the expected revenues amounted to 416,000 JD during the first year of the project and gradually increased to reach 542,786 JD during the tenth year. The period of capital redemption is 5 years with an internal rate of return equal to 21.35%, in addition for providing 28 jobs.

The following table shows the economic indicators of the project:

Table1 Economy indicators

Indicator	
Total Investment Cost (Jordan Dinars)	874.424
Internal Rate of Return (IRR)	21.35%
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	199.369
Job Opportunities provided by the Project	28
Total Revenues (For 10 Years)	4768974
Total Operating Expenses (For 10 years)	1539192
Total Net Profit (For 10 years)	1186327

Key Highlight of Jordan

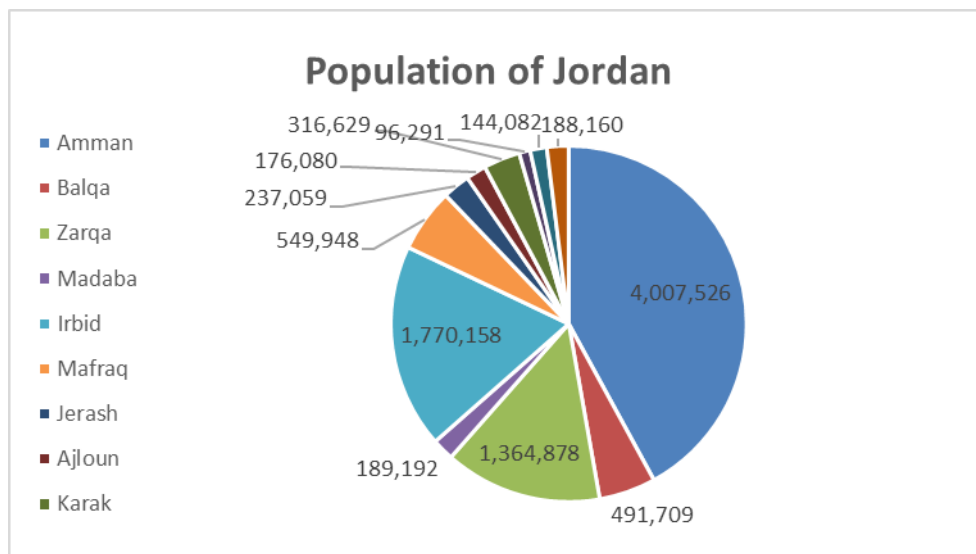
Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.



of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

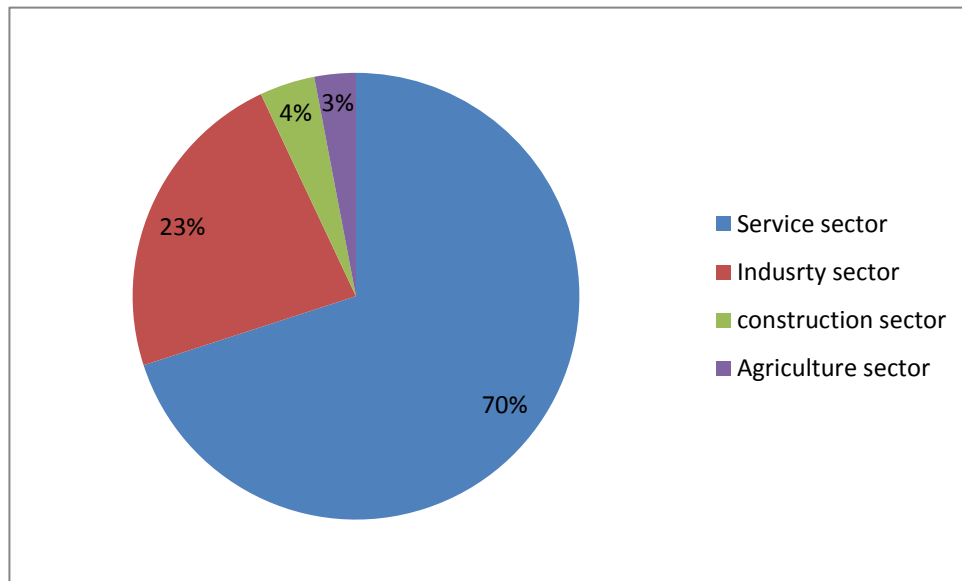
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Aqaba:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table3 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	Income Tax is 20% As per Investment Law categorization, Irbid is classified in Class (A) where investors enjoys 100% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.

2. Market Analysis

Project description

The project idea is to establish a children's Museum in Aqaba governorate, basically providing educational and recreational services for the citizens and visitors of Aqaba governorate particular below of 15 years in addition to other services. Aqaba governorate hasn't children museum for entertaining, and educational purposes in order to provide the child through various knowledge by entertainment method accepted and utilized by him.

The project is consisting of:

- **Art Studio:** Allocated Place to highlight and explore the artistic talents of children, through this place the children's museum designing activities and artistical programmed for all children of all ages to recognize each child his artistic side and artwork by joining various workshops and activities specifically designed to suit older ages.
- **Hall Exhibition:** the hall will be rich in educational exhibits which allow children to exercise their minds through interactive and educational activities. It is aimed to encourage the child to discover and not on the teleprompter.
- **Library:** it is helps to explore through the pleasure of reading of a large collection of books both Arabic and English language, address scientific topics.
- **Secret garden:** This space in the library will add green experience to the kids. The secret garden will provides green and natural place for visitors to connect with nature and learning about the environment and identify the quality of various plants which cultivated in Jordan.
- **Out door:** Deploy interactive exhibits in the outside area, an outdoor theater to revive reviews the various dramas outdoors. Some programs and exhibits will be held at the outside area in spring atmosphere to enhance the fun gameplay.
- **The human body:** The goal is educating children about human body organs and how they adapt to the environment and the most important functions of main parts.

- **Fly:** To give the children a full experience of flying as passengers or working in aviation field, this factor will boost the updated zone of knowledge to kids about everything related to aviation and allow them to interact with it using different tools, They can be pilots, flight engineers, passengers, airline and aircraft observer hosts.
- **Invention lab:** It provides space activities and educational programs enhance innate curiosity in children and develop their love of invention and rely on their ideas for creating inventions which can be kept by them and to be participated.
- **Car parking:** A place dedicated to lining up people's cars and visitors to the Museum.

Project objectives:

The objectives of the project:

- Achieving yield for the project owner.
- Governorate support in entertainment of children.
- Providing jobs for the citizens of the governorate.

Proposed service

Children's Museum project offers education and entertainment service for the region, in addition to birthdays and parties service.

Targeted segment

The targeted segments for the project are the families of the people of the region, visitors and tourists and for their children in particular.

Analysis of market size

According to the General Bureau of statistics of 2016, the populations of Jordanians at Aqaba governorate are 135,045 citizen and non-Jordanians population 53,114 individual, in terms of percentage of ages below 15 years nearly 37% according to the Department of statistics. The average number of foreign tourists to Aqaba governorate approximately 110,633 tourists during the last five years (2012-2016) addition to Jordanian visitors from different governorates which reached in holidays to 50,000 visitors.

Competitors analysis

There is an entertaining tutorial museum in Amman for children's, inside Al Hussein gardens which achieved remarkable success. Another Museum located in Ras Al Ain gallery serves the school and university students, there are rooms for seminars and activities as well. In addition to the Dead sea panorama Museum and the Museum of Jabber House in Salt.

Price analysis and pricing policy

Establish such project in the Aqaba region is a good and unique idea within the governorate in addition to create paradigm shift in the region's tourism sector. The income sources of the project will be based on member login the birthdays occasions in Museum as well as annual museum subscriptions, where the entry fee to Museum was estimated at (5) dinars per person and annual family subscription (200), JD, birthdays parties price per person (8) JD.

The following table shows the value of the expected prices of the Museum for 10 years inception:

Table 4 Expected prices during the period 2018-2024

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Entry fee	5.00	5.15	5.30	5.46	5.63	5.80	5.97	6.15	6.33	6.52
Annual family subscription	200	206	212	219	225	232	239	246	253	261
Price for hosting person/birthday	8	8	8	9	9	9	10	10	10	10

Market share

According to the General statistics of Aqaba region the number of population is 188,259 in habitants, where children under 15 year about 37% this means that there is almost 86,553 child, and the targeted customers for the museum is 48000 visitor/year. While the average rate of birthday parties about (400) part per year and the rate of attendances of birthday parties is (30).

Projected revenue

Projected annual revenues of the project

After the expected market share of the project have been calculated and determined to find annual price rate of the average museum, the following table shows the expected income for the project over the years (2018–2027).

Table5 Analyze revenue projections

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Entry fee	5.00	5.15	5.30	5.46	5.63	5.80	5.97	6.15	6.33	6.52
Annual family subscription	200	206	212	219	225	232	239	246	253	261
Price for hosting person/birthday	8	8	8	9	9	9	10	10	10	10
revenues of entry	240000	247200	254616	262254	270122	278226	286573	295170	304025	313146
The annual subscriptions revenues	80000	82400	84872	87418	90041	92742	95524	98390	101342	104382
Birthdays revenues	96000	98880	101846	104902	108049	111290	114629	118068	121610	125258
Total revenues	416000	428480	441334	454574	468212	482258	496726	511628	526976	542786

The revenues were calculated by multiplying the total average of the museum visitors by museum entry fees. The birthdays revenues, are calculated by summing the annual birthday rate by average attendance of visitors multiplied by hosting price per person, same way was used to calculate the annual subscriptions revenues.

3. Technical analysis

Technical feasibility study aims to develop technical specifications for the project, based on the marketing study results, in addition to assessing and reviewing the most important items of investment and operational costs. Technical project study includes: explaining and general description of the project and its components, in addition to structural and operational requirements for the project, and determine the general specifications to calculate the costs, according to the estimates the team work.

Human resources required

This project will need a general manager to supervise all the operational Museum aspects, accountant to manage transaction of the financial accounts of the Museum, in addition to the marketing side of the project. The following table shows the required employment details of the proposed project:

Table 6 Employment requirement

Job title	No.
General manager	1
Accountant	1
Marketing officer	1
Security & guarding	2
Monitor and control room	2
Total indirect	7
A receptionist	2
Tickets employee	2
Guidance staff	8
Parking worker	1
Chef	1
Waiter	2
Total direct	16
Total	23

- **General Manager:** propose the general policy of the Museum, the preparation of the special plans and execution follow-up, and technical and administrative supervision of staff, development of procedures, ensuring occupational health and safety requirements.
- **Accountant:** follow up the money aspects and the financial audit of the Museum.
- **Marketing officer:** Study the market needs and pursue publicity activities and assess its effectiveness, control the sales level to be regulated.
- **Security officer:** To maintain security and protecting visitors inside the Museum, and regulate the entry of visitors and any events carried out in the town.
- **Guard:** Monitoring the entry of visitors to the Museum, the closing and opening doors.
- **Monitor and control room:** Is responsible of the visitor's movement inside the Museum and control all aspects of the Museum through the cameras.
- **Receptionist:** Receive visitors, and ensure necessary access fees.
- **Tickets employee :** selling the tickets to visitors
- **Valet parking :** Lining up the cars of visitors inside the Museum's parking.

Site analysis

The project will be established in Aqaba which is characterized as existing on the Red Sea the beautiful climate all year round, in addition it is located in a convenient place to be targeted tourists. The project will be established in the city center, which will be overlooking the Red Sea and the Gulf of Aqaba.

Functional requirements analysis

Land and construction works

The land space required to build the project are 2000 m², the following table shows the land required for the project

Table7 the land required for the project

Land	Area(m2)
The land required	2,000

The construction work required for the project are as follows:

Table8 Construction works required

structural works	Area (m2)
Building & finishing work	750
Design	750
Excavation work	1,000
Land construction	500
Design and establish museum sections	250

Machinery and equipments

The following table shows the machinery and equipment needed for the project:

Table9 Machinery and equipment

List of machinery and equipments	No.
Fire fighting system	1
Central air conditioner	1

Furnitures

The project will be equipped with integrated office furniture to fulfill the Museum needs.

Legal required

- Title deed attached with the application.
- Last version of organization at design, attached with the application.
- Registration at the Ministry of industry and trade to be presented when final approval received.

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 13.57% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 10 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 11 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 12 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 13 Expenses Assumptions

Expenses Assumptions	
Utilities Expenses from Total Revenues	5%
Maintenance Expenses from Total Revenues	2.5%
Depreciation Expenses from Total Revenues	1%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	1%

Table 14 Income Tax Assumptions

Income Tax Assumptions	value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	0%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	Value
Annual Monthly Salaries have been calculated after multiplying monthly salaries by:(1)	16

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	%10.12
Income Tax Rate	5%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 15 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	6.50%
Return to Debt Maturity	9.48%
Market Risk Premuim	10.12%
Income Tax Rate	5%
Bets	1
Equity	542,655
Loans Value	349,770
Loans	
Cost of Borrowing before Tax	9.48%
Cost of Borrowing After Tax	9.01%
Loans Value	349,770
Loans Percentage	40%

Equity	
Cost of Equity	16.6%
Equity Value	524,655
Equity Percentage	60%
Gross	
Project Value	874,424
Weighted Average Cost of Capital	13.57%

Capital expenditures

The estimated cost of the project is 874,424 JD and covers land costs, construction works, machinery and equipment, furniture and furniture, pre-operating expenses, primary working capital. The following tables provide details of expenses:

Table 16 capital expenditure for the land

Land	Cost (JD/sq.m)	Area (sq.m)	Total Cost
The required Land	100	2000	200000
Total cost			200000

Table 17 capital expenditure of construction work

Construction work	Cost (JD/sq.m)	Area (sq.m)	Total
Buildings and finishing works	300	750	225,000
Design and blueprints	7	750	5,250
Digging	5	1000	5,000
Paving	12	500	6,000
Establishment of museum sections	750	500	187,500
Soil examination	200	-	200
Other	-	-	42,895
Grand Total			471,845

Table 18 expenses of machinery and equipment

Machinery and equipment	Quantity	Cost / JD	Total Cost
Fire-fighting system	1	5,000	5,000
Central air conditioner	1	15,000	15,000
Other	-	-	20,000
Total (JD)			40,000

Table 19 expenses of furniture and furnishings

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Office furniture	-	-	12,000
Total			12,000

Pre-operating expenses

Consist of a set of expenses that are expended before the start of the project and before achieving any revenue. Government fees, licenses, marketing expenses, training and transportation expenses are estimated as follows:

Table 20 pre-operationg expenses

Pre-operating expenses	Estemaited cost
Governmental fees	1,000
Transportation	750
marketing	3,500
Training	1,250
Total	6500

The initial working capital reflects the project's requirement to cover the costs incurred in operating the project. Until the project begins to pump income, the following table shows the components of the initial working capital:

Table 21 Share of Capital

Capital Share	No.	Percentage	Cost
Operating expenses	3	25%	32,384
General and administrative expenses	3	25%	2,962
Marketing expenses	3	25%	1,040
Salaries	3	25%	28,200
Total			64,586

Table 22 expenses summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	200,000	10%	220,000	25.2%
Construction works	471,845	10%	519,030	59.4%
Machinery and the equipment	40,000	10%	44,000	5.0%
Furniture	12,000	10%	13,200	1.5%
Pre-operating expenses	6,500	10%	7,150	0.8%
Working capital	64,586	10%	71,045	8.1%
Total (JD)	794,931		874,424	100%

Sources of funding

The financing structure of the project consists of loans and property rights. The investment cost of this project is estimated at (874,424) JD, where 40% of the project will be financed through loans by (349,770) JD, and the remaining (60%) through (Equity) with (524,655) JD. The following table shows the general structure of the required financing:

Table 23 sources of funding

Funding of Sources	Value	%
Property rights (self-financing)	524.655	60%
Loans	349.770	40%
Total	874.424	100%
The use of funding	Value	Ratio
Capital expenditure	796.230	91.06%
Pre-operating expenses	7150	0.82%
Working capital	71.045	8.12%
Total	874.424	100.00%

Operating expenses

The following tables show details of general operating, administrative and marketing costs as follows:

Table 24 Operating expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Service expenses	20,800	21,424	22,067	22,729	23,411	24,113	24,836	25,581	26,349	27,139
Maintenance expenses	10,400	10,712	11,033	11,364	11,705	12,056	12,418	12,791	13,174	13,570
Disposables	4,160	4,285	4,413	4,546	4,682	4,823	4,967	5,116	5,270	5,428
Direct salaries	82,400	85,778	89,295	92,956	96,768	100,735	104,865	109,165	113,640	118,300
Other expenses	11,776	12,220	12,681	13,160	13,657	14,173	14,709	15,265	15,843	16,444
Total	129,536	134,419	139,490	144,755	150,222	155,900	161,796	167,918	174,277	180,880

General and administrative expenses

The following table shows the expected administrative and general expenses of the proposed project (2018-2027):

Table 25 General and administrative expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	600	618	637	656	675	696	716	738	760	783
Stationary and Printing	600	618	637	656	675	696	716	738	760	783
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Training	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Insurance	5,972	6,151	6,335	6,525	6,721	6,923	7,131	7,344	7,565	7,792
Miscellaneous	1,077	1,109	1,143	1,177	1,212	1,249	1,286	1,325	1,365	1,405
Total	11,849	12,204	12,570	12,948	13,336	13,736	14,148	14,573	15,010	15,460

Marketing expenses

The following table shows the expected marketing expenses of the proposed project. Marketing expenses account for 1% of revenues based on the team's estimate:

Table 26 marketing expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total	4,160	4,285	4,413	4,546	4,682	4,823	4,967	5,116	5,270	5,428

Human resources

The salaries and monthly wages of each employee were estimated by 16 months for the purpose of calculating salaries and annual wages and their supplements from social security, health insurance, etc. to the administration staff. The following table shows the expected annual salaries and wages, where the rate of increase in salaries and wages per year was taken into account in the income statement

Table 27 expected staff numbers and wages (JD)

Job Title	No.	Monthly Salary	Sum of monthly salaries	Sum of yearly salaries
Non-Direct staff				
General Manager	1	1,500	1,500	24,000
Financial Manager	1	1,200	1,200	19,200
marketing manager	1	1,000	1,000	16,000
Accountant	2	400	800	12,800
Marketing officer	2	400	800	12,800
Guards	2	350	700	11,200
Monitor and Control	2	350	700	11,200
Maintenance	1	350	350	5,600
Total of non-direct	12		7,050	112,800
Receptionist	2	300	600	9,600
Tickets salesman	2	300	600	9,600
Museum staff	8	350	2,800	44,800
Nurse	1	400	400	6,400
Parking employee	1	250	250	4,000

Job Title	No.	Monthly Salary	Sum of monthly salaries	Sum of yearly salaries
Waiter	2	250	500	8,000
Total of direct	16		5,150	82,400
Total	28		12,200	195,200

The following table shows the expected annual salaries for employees for 10 years of the life of the project:

Table 28 expected annual salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-Direct staff										
General Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Financial Manager	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
marketing manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Accountant	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Marketing officer	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Security guards	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Monitorand Control	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Maintenance	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Total of non-direct	112,800	117,425	122,239	127,251	132,468	137,900	143,553	149,439	155,566	161,944
Direct staff										
Receptionist	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Tickets salesman	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Museum staff	44,800	46,637	48,549	50,539	52,612	54,769	57,014	59,352	61,785	64,318
Nurse	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Parking employee	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Waiter	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Total of direct	82,400	85,778	89,295	92,956	96,768	100,735	104,865	109,165	113,640	118,300
Total	195,200	203,203	211,535	220,207	229,236	238,635	248,419	258,604	269,207	280,244

Depreciations

The following tables summarize the cost of capital equipment, capital expenditure and consumption:

Table 29 Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation rates	Cost (JD)	Consumption rate	Percentage of annual
Land	220,000	%0.0	%0.0
Construction works	519,030	%5.0	%0.0
Machinery and equipment	44,000	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles and means of transportation	0	%10.0	%0.0
Gross Total (JOD)	796,230		

The table below summarizes capital costs, additions and annual depreciation:

Table 30 Depreciations and Additions on Construction Works

Capital costs and annual additions and consumption	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	796,230	797,770	799,360	801,003	802,70	804,456	806,27	808,145	810,08	812,091
Total Depreciation Expenses	31,671	31,825	31,985	32,149	32,319	32,494	32,676	32,863	33,057	33,258
Total Accumulated Depreciation	31,671	63,497	95,481	127,630	159,94	192,443	225,11	257,982	291,03	324,296
Total Additions	0	1,540	1,591	1,643	1,698	1,755	1,814	1,875	1,940	2,006
Total Net Book Values	764,558	734,273	703,879	673,373	642,75	612,013	581,15	550,164	519,04	487,795

Loans

The table below provides details of the loans and the repayment period of the project from the beginning until 2023.

Table 31 Loan Details

Loan	349,770
Annual Interest Rate	9.48%
Loan term	5
Loan Starts at year:	2018
Annual Payment	91,046
Number of Payments	5

The following table shows the loan payments:

Year	Yearly payments	Intrests	Capital	The remaining of loans
2018				349,770
2019	91,046	33,158	57,888	291,882
2020	91,046	27,670	63,375	228,507
2021	91,046	21,662	69,383	159,123
2022	91,046	15,085	75,961	83,162
2023	91,046	7,884	83,162	0

Income statement

All financial information and analysis of the financial expenses of the project shall be the estimated income statement for the years (2018-2027) as shown in the following table:

Table 32 comprehensive income statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	416,000	428,480	441,334	454,574	468,212	482,258	496,726	511,628	526,976	542,786
Gross Operating Revenues	416,000	428,480	441,334	454,574	468,212	482,258	496,726	511,628	526,976	542,786
Operating Expenses	(129,536)	(134,419)	(139,490)	(144,755)	(150,222)	(155,900)	(161,796)	(167,918)	(174,277)	(180,880)
Gross Operating Profit	286,464	294,061	301,845	309,820	317,989	326,358	334,930	343,709	352,700	361,905
<i>Gross Profit Percentage</i>	<i>%69</i>	<i>%69</i>	<i>%68</i>	<i>%68</i>	<i>%68</i>	<i>%68</i>	<i>%67</i>	<i>%67</i>	<i>%67</i>	<i>%67</i>
Salaries and Benefits (Indirect Staff)	(112,800)	(117,425)	(122,239)	(127,251)	(132,468)	(137,900)	(143,553)	(149,439)	(155,566)	(161,944)
General and Administraive Expenses	(11,849)	(12,204)	(12,570)	(12,948)	(13,336)	(13,736)	(14,148)	(14,573)	(15,010)	(15,460)
Markeing Expenses	(4,160)	(4,285)	(4,413)	(4,546)	(4,682)	(4,823)	(4,967)	(5,116)	(5,270)	(5,428)
Pre Operating Expenses	(7,150)									
Gross Indirect Expenses	(135,959)	(133,914)	(139,223)	(144,744)	(150,486)	(156,458)	(162,669)	(169,128)	(175,846)	(182,832)
Income before Interest, Depreciation and Tax	150,505	160,147	162,622	165,075	167,503	169,900	172,261	174,581	176,854	179,073
Fixed Assets Depreciations	(31,671)	(31,825)	(31,985)	(32,149)	(32,319)	(32,494)	(32,676)	(32,863)	(33,057)	(33,258)
Income before Tax and Interests	118,834	128,321	130,637	132,926	135,184	137,406	139,586	141,718	143,797	145,816
Bank Interests	(33,158)	(27,670)	(21,662)	(15,085)	(7,884)	(0)	(0)	(0)	(0)	(0)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	85,675	100,651	108,975	117,842	127,301	137,406	139,586	141,718	143,797	145,816
Income Tax	(4,284)	(5,033)	(5,449)	(5,892)	(6,365)	(6,870)	(6,979)	(7,086)	(7,190)	(7,291)
Net Profit	81,392	95,618	103,526	111,949	120,936	130,536	132,606	134,632	136,607	138,525
<i>Net Profit Percentage</i>	<i>%20</i>	<i>%22</i>	<i>%23</i>	<i>%25</i>	<i>%26</i>	<i>%27</i>	<i>%27</i>	<i>%26</i>	<i>%26</i>	<i>%26</i>
Compulsory Reserves	(8,139)	(9,562)	(10,353)	(11,195)	(12,094)	(13,054)	(13,261)	(13,463)	(13,661)	(13,852)
Retained Earnings	73,253	159,309	252,483	353,237	462,079	579,561	698,907	820,076	943,022	1,067,695

Expected cashflow statement

The following table shows the project's expected cash flow statement. Cash flows are positive for the length of the project, except for the first two years of construction (2018-2027).

Table 33 expected cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	81,392	95,618	103,526	111,949	120,936	130,536	132,606	134,632	136,607	138,525
Bank Interests	33,158	27,670	21,662	15,085	7,884	0	0	0	0	0
Depreciation	31,671	31,825	31,985	32,149	32,319	32,494	32,676	32,863	33,057	33,258
Total Operating Cashflows before Additions to Working Capital	146,221	155,114	157,173	159,183	161,138	163,030	165,282	167,495	169,664	171,782
Inventory (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	21,589	814	845	878	911	946	983	1,020	1,060	1,101
Working Capital (Increase/Decrease)	21,589	814	845	878	911	946	983	1,020	1,060	1,101
Net Cashflows from Operating Activities	167,811	155,928	158,018	160,061	162,049	163,976	166,265	168,516	170,724	172,883

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(796,230)	(1,540)	(1,591)	(1,643)	(1,698)	(1,755)	(1,814)	(1,875)	(1,940)	(2,006)
Net Cashflows from Investments Activities	(796,230)	(1,540)	(1,591)	(1,643)	(1,698)	(1,755)	(1,814)	(1,875)	(1,940)	(2,006)
Cashflows from Financing Activities										
Capital	524,655									
Loan Amortization	(57,888)	(63,375)	(69,383)	(75,961)	(83,162)	0	0	0	0	0
Bank Interest Rate	(33,158)	(27,670)	(21,662)	(15,085)	(7,884)	(0)	(0)	(0)	(0)	(0)
Loans	349,770	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	783,378	(91,046)	(91,046)	(91,046)	(91,046)	0	0	0	0	0
Net (Increase/Decrease) in Cash	154,960	63,342	65,382	67,372	69,305	162,221	164,451	166,640	168,784	170,877
Cashflows at the Beginning of Period	0	154,960	218,302	283,684	351,055	420,361	582,582	747,033	913,673	1,082,457
Cashflows at the End of Period	154,960	218,302	283,684	351,055	420,361	582,582	747,033	913,673	1,082,457	1,253,334

Expected balance sheet

A statement of financial position is one of the main financial statements on which the project is based. This list reflects the financial position of the entity and is usually assessed on the last day of the financial year. All financial information and analysis of the financial expenses of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 34 projected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	154,960	218,302	283,684	351,055	420,361	582,582	747,033	913,673	1,082,457	1,253,334
Inventory	0	0	0	0	0	0	0	0	0	0
Accounts Receivable	0	0	0	0	0	0	0	0	0	0
Total current Assets	154,960	218,302	283,684	351,055	420,361	582,582	747,033	913,673	1,082,457	1,253,334
Non Current Assets										
Fixed Assets (net)	764,558	734,273	703,879	673,373	642,752	612,013	581,151	550,164	519,046	487,795
Total Non Current	764,558	734,273	703,879	673,373	642,752	612,013	581,151	550,164	519,046	487,795
Assets										
Total Assets	919,518	952,575	987,562	1,024,428	1,063,113	1,194,595	1,328,184	1,463,837	1,601,503	1,741,129
Liabilities										
Current Liabilities										
Paybles	21,589	22,403	23,248	24,126	25,037	25,983	26,966	27,986	29,046	30,147

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Remaining amount of Loan	63,375	69,383	75,961	83,162	(0)	(0)	(0)	(0)	(0)	0
Total current Liabilities	84,965	91,787	99,209	107,288	25,037	25,983	26,966	27,986	29,046	30,147
Non Current Liabilities										
Long Terms Loans	228,507	159,123	83,162	0	0	0	0	0	0	0
Total Long Term Liabilities	228,507	159,123	83,162	0	0	0	0	0	0	0
Total Liabilities	313,471	250,910	182,371	107,288	25,037	25,983	26,966	27,986	29,046	30,147
Owners Equity										
Shareholders Contributions	524,655	524,655	524,655	524,655	524,655	524,655	524,655	524,655	524,655	524,655
Statutory Reseve	8,139	17,701	28,054	39,249	51,342	64,396	77,656	91,120	104,780	118,633
Retained Profits	73,253	159,309	252,483	353,237	462,079	579,561	698,907	820,076	943,022	1,067,695
Total Equity	606,046	701,665	805,191	917,140	1,038,076	1,168,611	1,301,218	1,435,850	1,572,457	1,710,982
Total Liabilities and Equity	919,518	952,575	987,562	1,024,428	1,063,113	1,194,595	1,328,184	1,463,837	1,601,503	1,741,129

Feasibility Indicators

In order to access the discounted cash flow statement, the discount factor must be calculated where more than one method is used to calculate it. However, in this study, WACC was used as a discount factor to calculate discounted cash flows and net value of investment. The Company's financing structure is based on equity and loans, and therefore the discount rate should be adjusted according to the WACC based on the following assumptions.

Based on the above, the following table shows the net expected free cash flow of the project

Table 35 Free Net Cash flows Table

Net cash flows	Pre-operation	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	The final value
Net Free Cashflows	(874,424)	174,961	154,388	156,427	158,418	160,351	162,221	164,451	166,640	168,784	170,877	1,856,115
Discount Factor	1.00	0.68	0.60	0.53	0.47	0.41	0.36	0.32	0.28	0.25	0.22	0.22
Net Present Value for Cash Flows	(874,424)	119,426	92,788	82,777	73,811	65,782	58,595	52,301	46,663	41,615	37,095	402,939

Summary of financial analysis indicators

The following table shows the recovery of capital for the project

Table 36 period of capital recovery

Capital payback period	Pre- Operation	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net free cash flow	(874,424)	174,961	154,388	156,427	158,418	160,351	162,221	164,451	166,640	168,784	170,877
Recovery	874,424	699,464	545,075	388,648	230,230	69,879	(92,342)	(256,793)	(423,433)	(592,217)	(763,094)
Payback period	5	1	1	1	1	1	0	0	0	0	0

Table 37 The results of the financial analysis

Indicator	Value
Discount Rate	13.57%
Net present value of cash flows	199,369
Capital payback period	5
Internal Rate of Return	21.35%

Sensitivity Analysis

The potential changes in the financial indicators were analyzed. Under this analysis, a framework for potential inputs was identified and the project's impact was measured by a decrease in revenues or a 10% increase in operating costs.

The following table shows that the project remains economically viable even if revenues fall by 10% or operating expenses increased by 10%.

Table 38 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback period	WACC
The original state	%21.35	5	13.57%
Decline in revenue 10%	%15.78	7	13.57%
Increase in operating expenses 10%	%19.43	5	13.57%

Conclusions

Based on the results of the financial analysis, and given that the internal rate of return of the project is higher than the discount rate, it was concluded that the project is profitable and economically feasible. The net present value of the project is positive by 199,369 JD. The project also provides 28 jobs for the region's residents.